

Multivariate statistics

Lecture 07:

Multivariate Multiple Linear Regression, MMLR

Literature and Problems: See course schedule

Agenda (preliminary):

SLIDE: Overview of linear regression models and terminology

Multivariate Multiple Linear Regression

- UMLR versus MMLR for vector responses, $Y = [Y_1 \dots Y_m]$
- model for response variable i : Y_i , $i = 1, \dots, m$
- observation j for response variable i : Y_{ji} , $i = 1, \dots, m$, $j = 1, \dots, n$
- n observations for m response variables on matrix form
- GLM, General Linear Model: $Y = Z\beta + \varepsilon$
- error vectors are independent ($\Rightarrow$ uncorrelated) between observations but the elements of each error vector can be correlated
- row-wise: $\varepsilon = \{ \varepsilon(j, :) \}_{j=1, \dots, n} \Rightarrow \text{Cov}[\varepsilon(j_1, :), \varepsilon(j_2, :)] = \delta_{j_1, j_2} \Sigma$, $\Sigma = \{\sigma_{pq}\}_{p, q=1, \dots, m}$
- column-wise: $\varepsilon = \{ \varepsilon(:, i) \}_{i=1, \dots, m} \Rightarrow \text{Cov}[\varepsilon(:, i_1), \varepsilon(:, i_2)] = \sigma_{i_1, i_2} I_{nn}$
- Least Squares estimation of $\beta(:, i) = \text{UMLR } \beta_{\text{hat}}$ for $Y_i \rightarrow$ total MMLR β_{hat}
- mean of β_{hat} and $\text{Cov}[\beta_{\text{hat}}(:, i_1), \beta_{\text{hat}}(:, i_2)]$
- mean and covariance matrix of ε_{hat}
- estimate of Σ , Σ_{hat}

MVN assumption for ε (for use in inference)

- $\varepsilon(j, :) \sim N_m(0, \Sigma) \Rightarrow$
- $\beta_{\text{hat}}(:, i) \sim N_{r+1}$ and $\varepsilon_{\text{hat}}^T \varepsilon_{\text{hat}} = n \Sigma_{\text{hat}, \text{ML}} = (n - (r+1)) \Sigma_{\text{hat}} \sim W_m(\Sigma, n - (r+1))$

Model check (analysis of residuals ε_{hat})

- univariate model check for $\varepsilon(:, i)_{i=1, \dots, m}$ as for UMLR
- MVN model check for $\varepsilon(j, :)_{j=1, \dots, n}$ with scatter plots + Mahalanobis distance

Inference on $\beta(:, i)$ as for UMLR

- CR, CI, ESS-based test for significant predictor variables

Simultaneous (true multivariate) ESS-based test for significant predictor variables

- $H_0: \beta_{(2)} = 0$, where $\beta = [\beta_{(1)}; \beta_{(2)}]$
- UMLR version: test-statistic $F = (\text{ESS}(\beta_{(1)} \rightarrow \beta) / \Delta \text{df}) / (\text{SS}_E / \text{df})$
- MMLR: $\varepsilon_{\text{hat}}^T \varepsilon_{\text{hat}}$ for full model and under $H_0 \rightarrow \text{ESS} = n \Sigma_{\text{hat}, \text{ML}, 1} - n \Sigma_{\text{hat}, \text{ML}}$
- LR test-statistic $\Lambda = (|\Sigma_{\text{hat}, \text{ML}}| / |\Sigma_{\text{hat}, \text{ML}, 1}|)^{n/2}$, reject H_0 for small Λ
- large sample approx.: $-(n - (r+1) - (m - r + q + 1)/2) \log(|\Sigma_{\text{hat}, \text{ML}}| / |\Sigma_{\text{hat}, \text{ML}, 1}|) \sim \chi^2_{m(r-q)}$

Predictions analogue to UMLR (equations in textbook)

MANOVA as a special case of MMLR analogue to UMLR (samme Z)